

Sequence Search with HMMER

27.05.2026

Computational Biotechnology, MOL.981UF

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What is HMMER?

- Homologous protein sequences provide the evolutionary information required for downstream alignment and ancestral sequence reconstruction.
- HMMER is a bioinformatics software suite based on Hidden Markov Models (HMMs) for sensitive sequence homology searching.
- A Hidden Markov Model is like a statistical fingerprint of a protein family that captures which amino acids are conserved, variable, inserted, or deleted during evolution.
- They are more sensitive than pairwise search methods

Why HMMER is powerful

- BLASTp mainly compares:
 - pairwise similarity
- HMMs model - entire protein family behavior, so they can detect:
 - weak evolutionary relationships
 - distant homologs
 - conserved motifs
 - even when similarity is low.

HMMER Web Server

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HMMER: biosequence analysis using profile hidden Markov models

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HMMER is used for searching sequence databases for sequence homologs, and for making sequence alignments. It implements methods using probabilistic models called profile hidden Markov models (profile HMMs).

HMMER is often used together with a profile database, such as [Pfam](#) or many of the databases that participate in [Interpro](#). But HMMER can also work with query *sequences*, not just profiles, just like BLAST. For example, you can search a protein query sequence against a database with **phmmer**, or do an iterative search with **jackhmmer**.

HMMER is designed to detect remote homologs as sensitively as possible, relying on the strength of its underlying probability models. In the past, this strength came at significant computational expense, but as of the new HMMER3 project, HMMER is now essentially as fast as BLAST.

HMMER can be downloaded and installed as a command line tool on your own hardware, and now it is also more widely accessible to the scientific community via [new search servers](#) at the European Bioinformatics Institute.

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EMBL-EBI

HMMER

Biosequence analysis using profile hidden Markov Models

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About HMMER

The HMMER web server: fast and sensitive homology searches. This site has been designed to provide near interactive searches for most queries, coupled with intuitive and interactive results visualisations.

PHMMER

EMBL-EBI

HMMER

Biosequence analysis using profile hidden Markov Models

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protein sequence vs protein sequence database

Sequence(s)

Paste in your sequence, use the [example](#), drag a file over or [choose a file to upload](#)

Submit

Reset

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JACKHMMER

EMBL-EBI

HMMER

Biosequence analysis using profile hidden Markov Models

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iterative search vs protein sequence database

Sequence/Alignment/Hmm

Paste in your alignment, hmm or sequence, use the [example](#), drag a file over or [choose a file to upload](#)

Submit

Reset

Clean

PHMMER vs JACKHMMER

Feature	PHMMER	JACKHMMER
Search type	Single-pass search	Iterative profile search
Algorithm	Query sequence vs Protein sequence database	Iterative search vs Protein sequence database
Sensitivity	Standard; finds close to moderately distant homologs	High; excels at finding extremely distant/remote homologs
Analogy	Comparable to a standard BLAST search	Comparable to PSI-BLAST
Speed	Very fast	Slower, requiring multiple passes to converge

Query sequence

>tr|A4Y0K1|A4Y0K1_ECTM1 Fatty acid desaturase
OS=Ectopseudomonas mendocina (strain ymp) OX=399739
GN=Pmen_4370 PE=4 SV=1

MSPSPASLNDQQRAAHIREQVMAHGNALRQRYPIHQDALGAGILAFALCG
MIGSAALYIGGHLPWWACL LNAFFASLT HEHDLIHS MYFRKQPLPHNLMLA
LVWLARPSTINPWVRRHLHLNHHKVSGSEADMEERAITNGEPWGIARLLMVG
DNMMSSFIRWLR AKNPEHRR LILTRTLKVYAPLG LLNWATWYLFLGFHLLDWA
AAALGAPIAWSASTLSVMQVVNVAVVVLVGPNVLR TFCLHFVSSNMHYYGDV
EPGNVIQQTQVLNPWWLWPLQAFCFNFGSSHAIH HFVVKEPFYIRQLTVPFAH
RVMREMGVRFNDFGTFARANRWTRRARTQQERASTA

PHMMER vs JACKHMMER

phmmer hmmscan hmmsearch jackhmmmer

protein sequence vs protein sequence database

Sequence(s)

Paste in your sequence, use the [example](#), drag a file over or [choose a file to upload](#)

Submit

Reset

Clean

Sequence database

UniProt (2025_01)

Taxonomy restriction

Enter your search terms or tax ID

Include all taxa

Clear all

Showing only taxa available in the selected database

Cut off

☒ E-value ☐ Bit Score

	Sequence	Hit
Significance E-values	0.01	0.03

phmmer hmmscan hmmsearch jackhmmmer

iterative search vs protein sequence database

Sequence/Alignment/Hmm

Paste in your alignment, hmm or sequence, use the [example](#), drag a file over or [choose a file to upload](#)

Submit

Reset

Clean

Sequence database

UniProt (2025_01)

Number of iterations

1

Taxonomy restriction

Enter your search terms or tax ID

Include all taxa

Clear all

Showing only taxa available in the selected database

PHMMER

[Search Details](#)[Customise Table](#)

Target	Description	Species	# Significant Hits	E-value
> A4Y0K1	Fatty acid desaturase	Ectopseudomonas mendocina	1	5.1e-243
> A0A1H0TZJ9	Fatty acid desaturase	Ectopseudomonas guguanensis	1	2.0e-234
> A0A5N7XUL6	Fatty acid desaturase	Pseudomonas sp.	1	2.0e-234
> A0A1G7NXB6	Fatty acid desaturase	Ectopseudomonas alcaliphila	1	5.9e-233
> A0A1I5YIW9	Fatty acid desaturase	Ectopseudomonas toyotomiensis	1	6.7e-232
> A0AA42IPP9	Fatty acid desaturase	Ectopseudomonas toyotomiensis	1	1.3e-231
> A0A7Y1FEW4	Fatty acid desaturase	Pseudomonas sp. WS 5019	1	3.6e-231
> A0A1H2ML97	Fatty acid desaturase	Pseudomonas sihuiensis	1	9.1e-231
> A0A328VP58	Fatty acid desaturase	Pseudomonas sp. MDMC224	1	9.7e-230
> A0A1L7G6A9	Fatty acid desaturase	Ectopseudomonas alcaliphila	1	1.3e-229
> A0A379IP61	Fatty acid desaturase	Ectopseudomonas mendocina	1	2.8e-229
> A0A550G1W6	Fatty acid desaturase	Pseudomonas sp. ALS1131	1	3.7e-229
> A0A3D0JN19	Fatty acid desaturase	Pseudomonas sp.	1	5.2e-229
> A0A1G6W1A2	Fatty acid desaturase	Ectopseudomonas chengduensis	1	8.8e-229

PHMMER Results

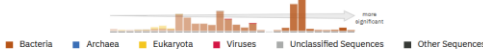
Browse Phmmer Results

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Sequence Features and Matches

FA_desaturase

Distribution of Significant Hits

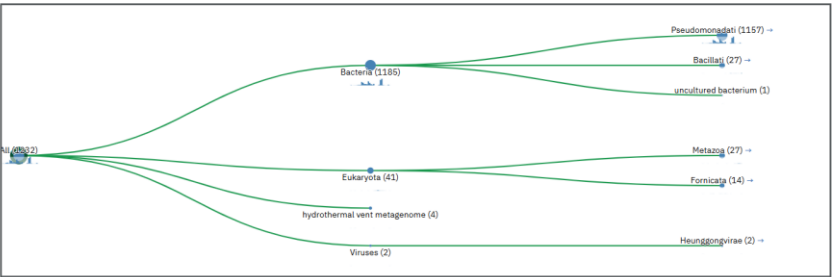


Search Details

Customise Table

Target	Description	Species	# Significant Hits	E-value
> A4Y0K1	Fatty acid desaturase	Ectopseudomonas mendocina	1	5.1e-243
> A0A1H0TZJ9	Fatty acid desaturase	Ectopseudomonas guguanensis	1	2.0e-234
> A0A5N7XUL6	Fatty acid desaturase	Pseudomonas sp.	1	2.0e-234
> A0A1G7NXB6	Fatty acid desaturase	Ectopseudomonas alcaliphila	1	5.9e-233
> A0A115YIW9	Fatty acid desaturase	Ectopseudomonas toyotomiensis	1	6.7e-232
> A0AA42IPP9	Fatty acid desaturase	Ectopseudomonas toyotomiensis	1	1.3e-231

Reset zoom



Use Ctrl + Scroll to zoom, drag to move around.

Lineage: → All

All (1232)

Species	Count	View
Pseudomonas fluorescens	69	Show
Acinetobacter baumannii	16	Show
Pseudomonas syringae group genomsp. 3	15	Show

[Score](#) [Taxonomy](#) [Domain](#) [Download](#)

Domain Architecture	Count
> FA_desaturase example: A4Y0K1	890 View Scores
> example: A0AAN0PAW9	226 View Scores
> FA_desaturase x2 example: A0A115YIW9	115 View Scores
> FA_desaturase, FAD_binding_6, Fer2 example: A0A965ZPF7	1 View Scores

```
phmmer results for job 0451b22c-0fc8-4044-9367-f585639a5b2d:

Target Num-hits Bias Bit-Score E-value Tax-Id Species Description
-----
A4Y0K1_ECTM1 1 17.2 819.1 5.10e-243 300 Ectopseudomonas mendocina Fatty acid desaturase

Target-env-start Target-env-end Target-ali-start Target-ali-end Query-start Query-end E-value(cond.)
-----
1 352 1 352 1 352 3.00e-248

QUERY
mspspaslndqraahireqvmahgnalrqrypilqhodalggagilafalcmigsaalyigghlpwacillnaffasltlehdlihsmyfrkqplphnlmlalvlarpstinpvrhhlhnhkvsgeadmeeraaitngepwiarlml
vgdnnmssfiwlraknephrrlilrtlkvyaplgllnwatwylfifghlldwaaalgapiaawsastlsvmqvvnvavvlgpvnrltfcflhfvsnnhyygdvpgnviqqtqvlnpwlpqlaefcnfgsshaihhfvvkepfyirqltv
pfahrvmremgvrnfndgtfaranrwtrrrartqgerasta

MATCH
mspspaslndqraahireqvmahgnalrqrypilqhodalggagilafalcmigsaalyigghlpwacillnaffasltlehdlihsmyfrkqplphnlmlalvlarpstinpvrhhlhnhkvsgeadmeeraaitngepwiarlml
vgdnnmssfiwlraknephrrlilrtlkvyaplgllnwatwylfifghlldwaaalgapiaawsastlsvmqvvnvavvlgpvnrltfcflhfvsnnhyygdvpgnviqqtqvlnpwlpqlaefcnfgsshaihhfvvkepfyirqltv
pfahrvmremgvrnfndgtfaranrwtrrrartqgerasta
PP 89
*****
*****97*****

TARGET
MSPSPASLNDQRAAHIREQVMAHGNALRQRYPILOQDQALGAGILAFALCMIGSAALYIGGHLPWACILLNAFFASLTLEHDLIHSYFRKQPLPHNLMALVWLARPSTINPVRRLHLNHHKVSSEADMEERAITNGEPWGIARLLM
VGDNNMSSFIWLRKNEPHERRLILRTLKVYAPLGLLNWATWYLFIFGHLLDWAALGAPIAWASTLSVMQVNVAVVVLGPNVLRFTFCLHFVSSNMHYYGDEPGNVIQQTQVLNPWLPLQAFCFNFGSSHAHHFVVKPEFYIRQLTV
PFHRVVRMGVRFNDGTGFARANRWTRRRARTQGERASTA

-----
A0A1H0TZJ9_9GAWW 1 17.0 790.8 1.99e-234 1198456 Ectopseudomonas guguanensis Fatty acid desaturase

Target-env-start Target-env-end Target-ali-start Target-ali-end Query-start Query-end E-value(cond.)
-----
1 352 1 351 1 351 1.17e-239

QUERY
mspspaslndqraahireqvmahgnalrqrypilqhodalggagilafalcmigsaalyigghlpwacillnaffasltlehdlihsmyfrkqplphnlmlalvlarpstinpvrhhlhnhkvsgeadmeeraaitngepwiarlml
vgdnnmssfiwlraknephrrlilrtlkvyaplgllnwatwylfifghlldwaaalgapiaawsastlsvmqvvnvavvlgpvnrltfcflhfvsnnhyygdvpgnviqqtqvlnpwlpqlaefcnfgsshaihhfvvkepfyirqltv
pfahrvmremgvrnfndgtfaranrwtrrrartqgerast
```

JACKHMMER

Cut off

☒ E-value

☐ Bit Score

Significance E-values

Sequence

0.001

Hit

0.03

Report E-values

Sequence

1

Hit

1

Jackhmmer Summary

			Hits			
Iteration	Status	Results	New	Lost	Dropped	Total
1	SUCCESS	dc4598e7-8d57-4dd5-a7df-81a91988b728	+1232	-	-	1232
2	SUCCESS	56cd1a0f-bc77-4110-9903-a78cc9ade21d	+733	-	-	1965
3	SUCCESS	d1bd77f8-7090-418b-a908-16d5cb8f8d3c	+38241	-	-	40206

Run next iteration

Profile drift.

3 iterations

Cut off

☒ E-value ☐ Bit Score

Significance E-values

Sequence

0.00001

Hit

0.03

Report E-values

Sequence

1

Hit

1

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Jackhmmer Summary

			Hits			
Iteration	Status	Results	New	Lost	Dropped	Total
1	SUCCESS	4f1e3ea0-7ee8-49b6-bfc0-a8409e6ca7af	+1228	-	-	1228
2	SUCCESS	b1fe607f-4d3b-4621-bfe8-08cdb5ceb97c	+61	-	-	1289
3	SUCCESS	59a47f6e-dbc8-464c-9762-98c6e757b648	+2504	-	-	3793

3 iterations

Search Details		Customise Table		
Target	Description	Species	E-value	
> A4Y0K1	Fatty acid desaturase	Ectopseudomonas mendocina	5.1e-243	✓
> A0A1H0TZJ9	Fatty acid desaturase	Ectopseudomonas guguanensis	2.0e-234	✓
> A0A5N7XUL6	Fatty acid desaturase	Pseudomonas sp.	2.0e-234	✓
> A0A1G7NXB6	Fatty acid desaturase	Ectopseudomonas alcaliphila	5.9e-233	✓
> A0A1I5YIW9	Fatty acid desaturase	Ectopseudomonas toyotomiensis	6.6e-232	✓
> A0AA42IPP9	Fatty acid desaturase	Ectopseudomonas toyotomiensis	1.3e-231	✓
> A0A7Y4EEW4	Fatty acid desaturase	Pseudomonas sp. WS 5018	2.6e-231	✓

Search Details		Customise Table		
Target	Description	Species	E-value	
> A0A7Y6EB53	Fatty acid desaturase	Pseudomonas sp. C2B4	8.5e-176	✓
> A4Y0K1	Fatty acid desaturase	Ectopseudomonas mendocina	9.4e-176	✓
> A0A5E7WA35	Fatty acid desaturase domain-containing protein	Pseudomonas fluorescens	2.8e-175	✓
> A0A3T0JMH8	Fatty acid desaturase	Pseudomonas syringae	3.1e-175	✓
> A0A554PKI5	Fatty acid desaturase	Pseudomonas sp. ATCC 13985	3.4e-175	✓
> A0A6L5BY50	Fatty acid desaturase domain-containing protein	Pseudomonas	3.5e-175	✓

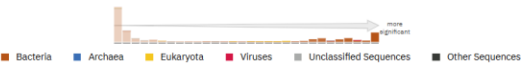
Search Details		Customise Table		
Target	Description	Species	E-value	
> A0A7Z1GN24	Fatty acid desaturase	Pseudomonas poae	1.4e-168	✓
> A0A7Y6EB53	Fatty acid desaturase	Pseudomonas sp. C2B4	2.3e-168	✓
> A0A7Y5YJR0	Fatty acid desaturase	Pseudomonas sp. C1C7	2.3e-168	✓
> A0A263NT71	Fatty acid desaturase	Pseudomonas sp. IB20	2.7e-168	✓
> A0A0D0SCG3	Fatty acid desaturase	Pseudomonas fluorescens	3.0e-168	✓
> A0A1X7H133	Fatty acid desaturase	Pseudomonas sp. LAM017WK12:11	3.8e-168	✓
> A0A5E7WA35	Fatty acid desaturase domain-containing protein	Pseudomonas fluorescens	4.1e-168	✓
> A0A7Y8DRN5	Fatty acid desaturase	Pseudomonas tolaasii	4.7e-168	✓

JACKHMMER Results

Sequence selection

☒ above threshold ☐ unselect all

Distribution of Significant Hits



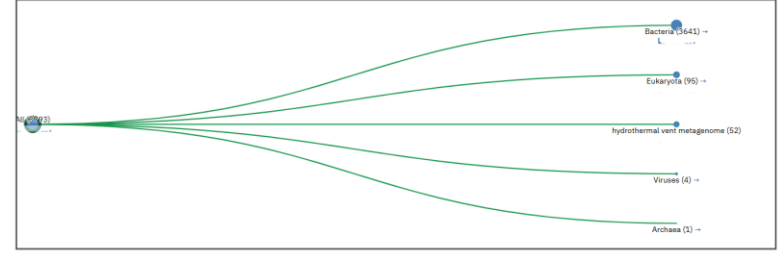
Search Details

Customise Table

Target	Description	Species	E-value	
> A0A7Z1GN24	Fatty acid desaturase	Pseudomonas poae	1.4e-168	✓
> A0A7Y6EB53	Fatty acid desaturase	Pseudomonas sp. C2B4	2.3e-168	✓
> A0A7Y5YJR0	Fatty acid desaturase	Pseudomonas sp. C1C7	2.3e-168	✓
> A0A263NT71	Fatty acid desaturase	Pseudomonas sp. IB20	2.7e-168	✓
> A0A0D0SCG3	Fatty acid desaturase	Pseudomonas fluorescens	3.0e-168	✓
> A0A1X7H133	Fatty acid desaturase	Pseudomonas sp. LAM017WK12:I1	3.8e-168	✓
> A0A5E7WA35	Fatty acid desaturase domain-containing protein	Pseudomonas fluorescens	4.1e-168	✓
> A0A7Y8DRN5	Fatty acid desaturase	Pseudomonas tolaasii	4.7e-168	✓
> A0A5P1DG25	Fatty acid desaturase	Pseudomonas haemolytica	4.7e-168	✓
> A0A6A6N7R4	Fatty acid desaturase	Pseudomonas haemolytica	4.9e-168	✓

Score Taxonomy Domain Download

Reset zoom



Use Ctrl + Scroll to zoom, drag to move around.

Lineage: → All

All (3793)

Species	Count	View
Gammaproteobacteria bacterium	80	Show
Pseudomonas fluorescens	73	Show
Mesorhizobium sp.	66	Show

Score Taxonomy Domain Download

Domain Architecture	Count	
FA_desaturase example: A0A949P3H7	2702	View Scores
FA_desaturase x2 example: A0A7Z7IRW2	504	View Scores
example: A0A446ZMU9	227	View Scores
FA_desaturase, FAD_binding_6, NAD_binding_1, Fer2 example: A0A238U830	118	View Scores
FA_desaturase x3 example: A0A841WX90	51	View Scores

```
jackhammer results for job 59a47f6e-dbc8-464c-9762-98c6e75b648:

=====
Target Num-hits Bias Bit-Score E-value Tax-Id Species Description
=====
A0A7Z1GN24_9PSED 1 10.2 573.9 1.43e-168 200451 Pseudomonas poae Fatty acid desaturase
=====
Target-env-start Target-env-end Target-ali-start Target-ali-end Query-start Query-end E-value(cond.)
10 345 11 342 2 333 1.30e-172
QUERY
ltdekkikkirkaikaeeeklrkrypiikhqdaigllgillslagmlsavlylaglipawlaillnAflaslhlheHDIHslyfkkqkVhnlmllwlarpstinpvrnk1HlHhHkvsGteeDleerlitnGekwgl1rl1lvldnlls
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lwlarpstinpvr++rr1HlHhHkvsGteeDleerlitnGekwgl1rl1lvldnlls++k+sl+ 1+++plall+++wylf1f+hn++asllg+ii+sg
+1++++v++1+avv+i+PnvlR+fcLhfvsnnmHYgdve+gnv+qqtqvl+wnl+Plq+fcfnfstH+IHfVvkePFYirq+tak ahkv++e+GvrfNDfgtfrknaNrye
PP 899
=====
*****86
=====
TARGET
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AFIRMLRARTWGHKLKLRSLLVAPLALLHGAWYFLFGHAANGIASLGTPIDNSAGTLQWQVQIDIAAVVIIGPNVLRFLCHLFVSSNHYYGDEPGNVIQQTQVLNPNWLLPLQACFCNFGSTHG1HfVvkePFYirqltakkahkv1
AEMGVRFNDFGTFRANRLE
=====
A0A7Y6EB53_9PSED 1 8.5 573.2 2.25e-168 2735270 Pseudomonas sp. C2B4 Fatty acid desaturase
=====
Target-env-start Target-env-end Target-ali-start Target-ali-end Query-start Query-end E-value(cond.)
10 345 11 344 2 335 2.06e-172
QUERY
ltdekkikkirkaikaeeeklrkrypiikhqdaigllgillslagmlsavlylaglipawlaillnAflaslhlheHDIHslyfkkqkVhnlmllwlarpstinpvrnk1HlHhHkvsGteeDleerlitnGekwgl1rl1lvldnlls
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reaGvrfNDfgtfrknaNrye
MATCH 1+++++lr++ ae +lr+rp+pl+hqda+g+gil++slgmi+sa+ly++g++p++lln+f+asl+hleHDIHs+yf+kq++ hn1mm
```


Thank You